Experiment-5

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April 16, 2026

Contents

1 Title

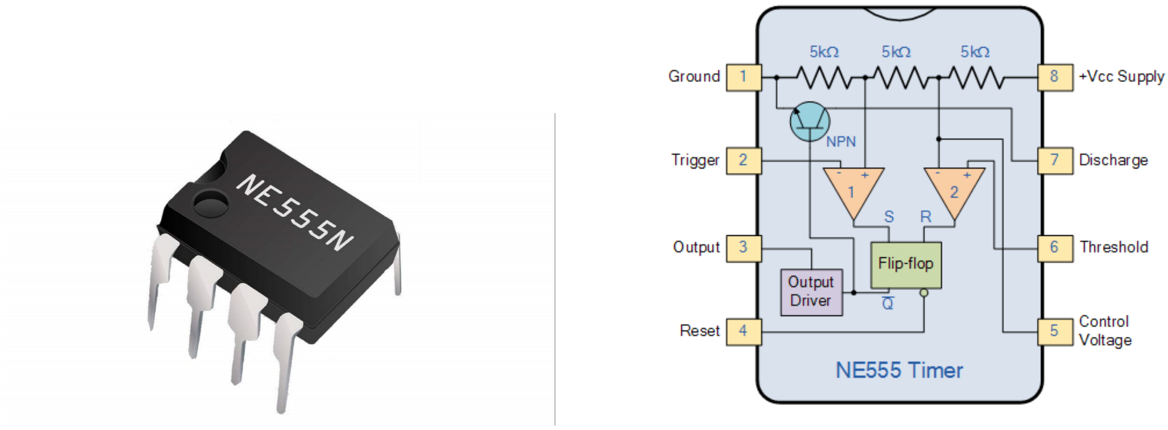
Study of 555 Timer IC as Astable Multivibrator with Different Frequencies.

2 Aim

The objective of this experiment is to study the 555 timer IC in astable mode and observe how its output waveform changes with different timing resistor and capacitor values. Using measured values of R_A, R_B, and C, we compare experimental timing parameters (T_m, T_s, and f) with their theoretical expressions, and check how well the expected RC-dependent behaviour is reproduced in practice.

3 Theory

3.1 The 555 Timer IC



(a) NE555 timer IC package (SMD).

(b) Pin configuration of the NE555 timer.

Figure 1: NE555 IC and its pinout used for circuit understanding and wiring.

The 555 timer is one of the most widely used ICs in basic electronics because it can generate accurate delays and oscillations with very little external circuitry. Depending on how it is connected, it can work as a monostable pulse generator, astable oscillator, or bistable element. In this experiment we use the astable configuration.

Internally, the IC contains:

- two voltage comparators (C_1 and C_2),
- an SR latch (flip-flop),
- a discharge transistor (Q_1), and
- a three-resistor divider that creates reference levels at $\frac{1}{3}V_{CC}$ and $\frac{2}{3}V_{CC}$.

The basic switching logic is simple:

- if the threshold pin rises above $\frac{2}{3}V_{CC}$, the latch resets and output goes LOW,
- if the trigger pin falls below $\frac{1}{3}V_{CC}$, the latch sets and output goes HIGH.

This repeated set–reset action drives periodic charging and discharging of the timing capacitor.

3.2 Astable Operation

In astable mode, the 555 has no stable rest state; it continuously toggles between HIGH and LOW. Therefore, it produces a continuous rectangular waveform at the output.

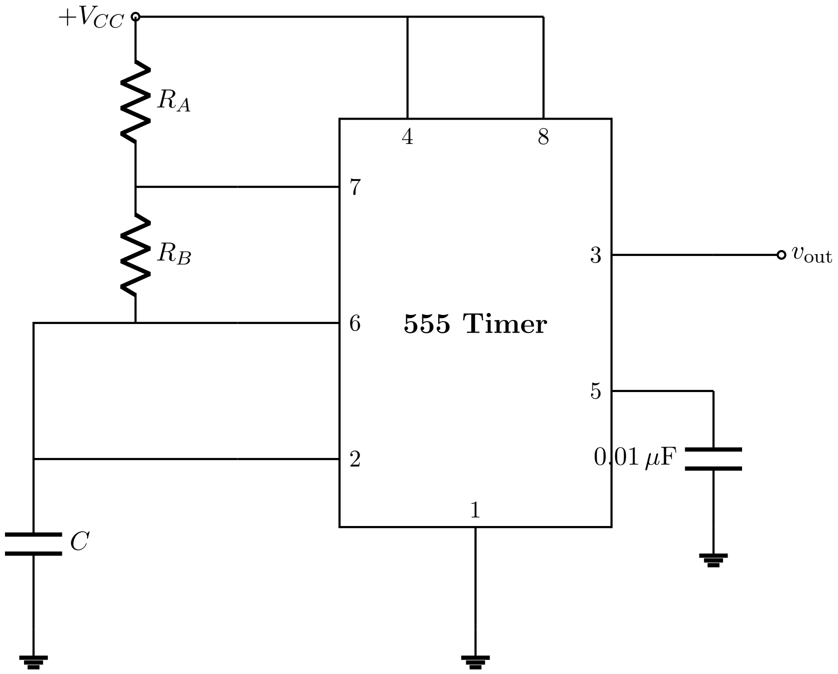


Figure 2: IC 555 connected as an astable multivibrator.

The capacitor voltage v_c moves repeatedly between the two internal thresholds: $\frac{1}{3}V_{CC}$ and $\frac{2}{3}V_{CC}$.

- When output is HIGH, the discharge transistor is OFF, and the capacitor charges through $(R_A + R_B)$.
- When v_c reaches $\frac{2}{3}V_{CC}$, output switches LOW, the discharge transistor turns ON, and the capacitor discharges through R_B.
- When v_c falls to $\frac{1}{3}V_{CC}$, output switches HIGH again, and the cycle repeats.

Hence, the output is a square/rectangular wave, while v_c is an exponential charge–discharge waveform.

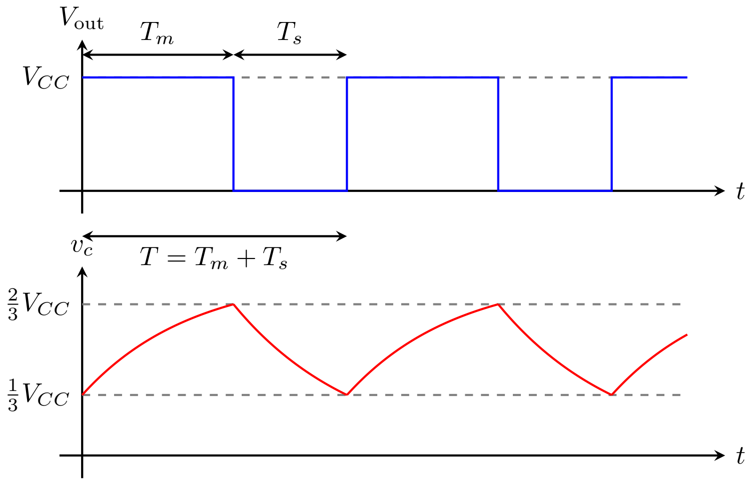


Figure 3: Idealised waveforms: (top) timer output V_{out}; (bottom) capacitor voltage v_c.

3.3 Derivation of Time Period and Frequency

Charging interval (T_m, mark time). During the HIGH state, the capacitor charges from $\frac{1}{3}V_{CC}$ to $\frac{2}{3}V_{CC}$ through $R_A + R_B$:

$$v_c(t) = V_{CC} - \frac{2}{3}V_{CC}\, e^{-t/[(R_A+R_B)C]}.$$

Applying the upper threshold condition at $t = T_m$:

$$\boxed{T_m = C(R_A + R_B)\ln 2 \approx 0.693\, C(R_A + R_B)}. \tag{1}$$

Discharging interval (T_s, space time). During the LOW state, the capacitor discharges from $\frac{2}{3}V_{CC}$ to $\frac{1}{3}V_{CC}$ through R_B:

$$v_c(t) = \frac{2}{3}V_{CC}\, e^{-t/(R_B C)}.$$

Applying the lower threshold condition at $t = T_s$:

$$\boxed{T_s = CR_B \ln 2 \approx 0.693\, CR_B}. \tag{2}$$

Total period and frequency.

$$T = T_m + T_s = C(R_A + 2R_B)\ln 2 \approx 0.693\, C(R_A + 2R_B), \tag{3}$$

$$f_{\text{Th}} = \frac{1}{T} \approx \frac{1.44}{C(R_A + 2R_B)}. \tag{4}$$

The duty cycle is:

$$D = \frac{T_m}{T} = \frac{R_A + R_B}{R_A + 2R_B}. \tag{5}$$

So for $R_B \gg R_A$, the duty cycle approaches 50%.

4 Experimental Setup

4.1 Components Used

- DC power supply: +15 V

- IC 555 (NE555), 1 unit
- Resistors R_A: 1.0 kΩ, 9.8 kΩ, 101 kΩ (measured values)
- Resistors R_B: 9.8 kΩ, 101 kΩ, 0.98 MΩ (measured values)
- Capacitors C: 0.001 μF, 0.01 μF, 0.1 μF, 1 μF, 10 μF
- Breadboard, single-strand wires
- Cathode Ray Oscilloscope (CRO)
- LED (for visual observation at low frequencies)

5 Data and Observations

Theoretical values of T_m, T_s, and f_{Th} were calculated using Equations (1)–(4). For better accuracy, measured resistor values were used in calculation ($R_A \in \{1.0, 9.8, 101\}$ kΩ, $R_B \in \{9.8, 101, 980\}$ kΩ).

The percentage error in frequency is defined as:

$$\%\,\text{Error} = \frac{|f_{\text{Th}} - f_{\text{exp}}|}{f_{\text{Th}}} \times 100.$$

Table 1: Comparison of theoretical and experimental results for all RC configurations.

Sl. No.	C (μF)	R_A (kΩ)	R_B (kΩ)	f_{Th} (Hz)	T_m^{Th} (ms)	T_s^{Th} (ms)	T_m^{Exp} (ms)	T_s^{Exp} (ms)	f_{exp} (Hz)	% Error
1	0.001	1.0	9.8	70033.7	0.00749	0.00679	0.010	0.010	50000.0	28.61
2	0.001	9.8	101.0	6811.6	0.07680	0.07001	0.092	0.100	5208.3	23.54
3	0.001	101.0	980.0	700.0	0.74929	0.67928	0.890	0.980	534.8	23.61
4	0.01	1.0	9.8	7003.4	0.07486	0.06793	0.089	0.100	5291.0	24.45
5	0.01	9.8	101.0	681.2	0.76801	0.70008	0.920	1.000	520.8	23.54
6	0.01	101.0	980.0	70.0	7.49292	6.79284	9.500	10.40	50.3	28.21
7	0.1	1.0	9.8	700.3	0.74860	0.67928	0.630	0.750	724.6	3.47
8	0.1	9.8	101.0	68.1	7.68007	7.00079	6.700	7.900	68.5	0.55
9	0.1	101.0	980.0	7.0	74.9292	67.9284	66.00	78.00	6.9	0.79
10	1	1.0	9.8	70.0	7.48599	6.79284	8.20	9.20	57.5	17.94
11	1	9.8	101.0	6.8	76.8007	70.0079	86.0	94.0	5.6	18.44
12	1	101.0	980.0	0.7	749.292	679.284	860.0	920.0	0.6	19.74
13	10	1.0	9.8	7.0	74.8599	67.9284	86.0	96.0	5.5	21.54
14	10	9.8	101.0	0.7	768.007	700.079	920.0	990.0	0.5	23.14
15	10	101.0	980.0	0.07	7492.92	6792.84	9100.	9400.	0.05	22.78

6 Analysis and Result

6.1 Overall Agreement

The experimental trends follow theory well, though not perfectly. Across all fifteen observations, the mean frequency error is about **18.69%**. Given that measured resistor values were already used in the theoretical calculation, the remaining mismatch is mainly due to capacitor non-idealities and practical circuit effects.

6.2 Dependence on Frequency Range

A clear pattern appears in the data:

- **Best agreement** occurs in the mid-frequency range ($C = 0.1\,\mu$F, rows 7–9), where error is only 0.55% to 3.47%.
- **Larger error** appears at both extremes: high frequency ($C \leq 0.01\,\mu$F, rows 1–6) and very low frequency ($C \geq 1\,\mu$F, rows 10–15).

At high frequency, the period is very small, so internal switching delay of the 555 and stray breadboard capacitance become non-negligible compared to T, reducing f_{exp}. At low frequency, electrolytic capacitor tolerance and leakage become dominant and increase the measured period.

6.3 RC Scaling Check

The expected scaling law $T \propto RC$ is observed clearly. For fixed resistor values, increasing C by a factor of 10 increases measured T_m and T_s by about 10 and decreases f_{exp} by about 10. A similar trend is seen when the resistor pair is increased.

6.4 Relation Between T_m and T_s

In all cases, $T_m > T_s$, which is exactly what the astable model predicts: charging happens through $(R_A + R_B)$, while discharging occurs through R_B only. The theoretical ratio

$$\frac{T_m}{T_s} = \frac{R_A + R_B}{R_B}$$

is close to 1.10 for the resistor sets used, and the measured values are consistent with this.

7 Sources of Error

- **Capacitor tolerance and leakage (major at low frequency):** especially for electrolytic capacitors ($1\,\mu$F and above), actual value can be significantly higher than nominal, and leakage distorts ideal exponential behaviour.
- **Internal non-ideal behaviour of the NE555:** finite propagation delay and slight deviation of comparator thresholds from ideal $\frac{1}{3}V_{CC}$, $\frac{2}{3}V_{CC}$ values.
- **Breadboard parasitics:** stray capacitance and small contact resistances modify effective R and C, especially when timing capacitor is very small.
- **CRO reading limitations:** cursor placement error, finite sampling resolution, and timebase accuracy all contribute to uncertainty in T_m and T_s.
- **Supply fluctuations and wiring noise:** small variations in V_{CC} shift threshold levels and can slightly alter timing.

8 Conclusion

The 555 timer was successfully operated as an astable multivibrator over a wide frequency range. By varying R_A, R_B, and C, stable periodic output waveforms were obtained from about 50 kHz down to roughly 0.05 Hz.

The measured timing and frequency values show good qualitative agreement with the standard astable expressions, and very good quantitative agreement in the mid-frequency region.

Higher deviations at very high and very low frequencies are consistent with known practical effects (propagation delay, parasitic elements, and capacitor non-idealities).

Overall, the experiment confirms the expected dependence of period and frequency on the RC network and validates the 555 astable model within normal laboratory uncertainty.